

Convolutional Neural Network (CNN)

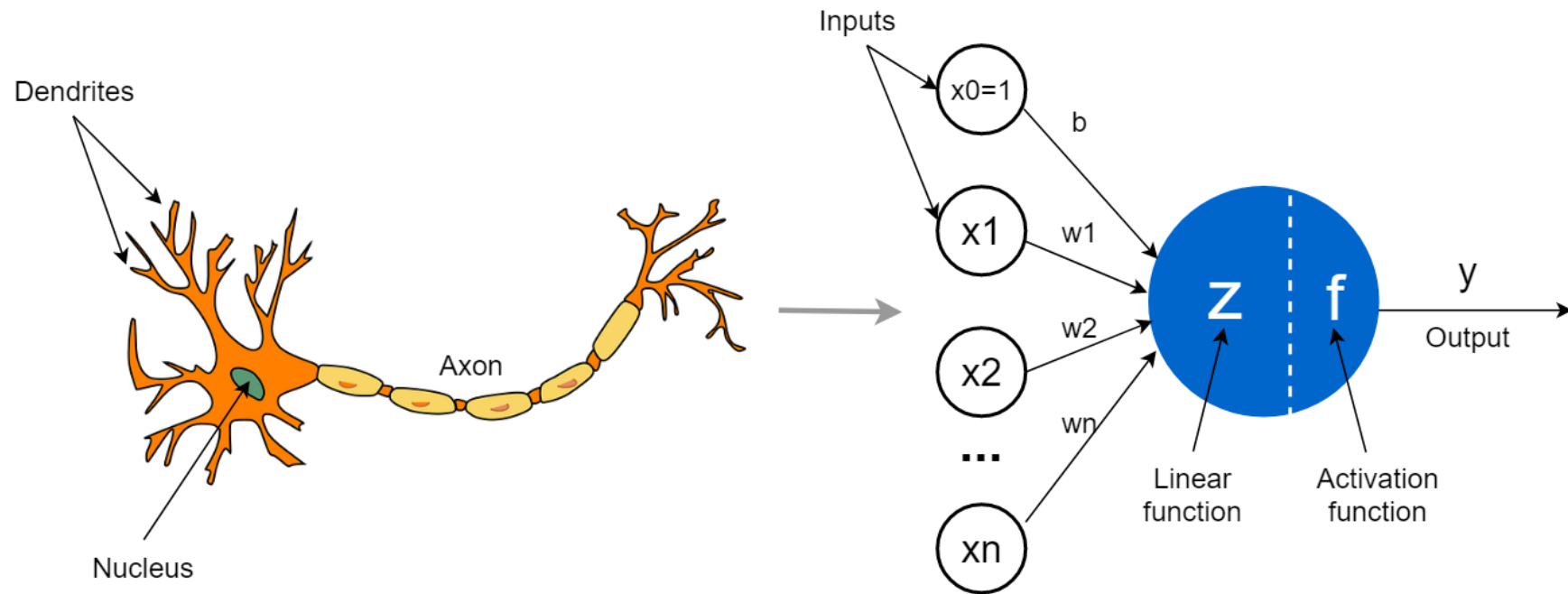
김우주

2026.05.23

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- 2주차 복습
- Multi-Layer Perceptron의 한계
- Convolution 연산
- Convolutional Neural Network의 구조

Perceptron

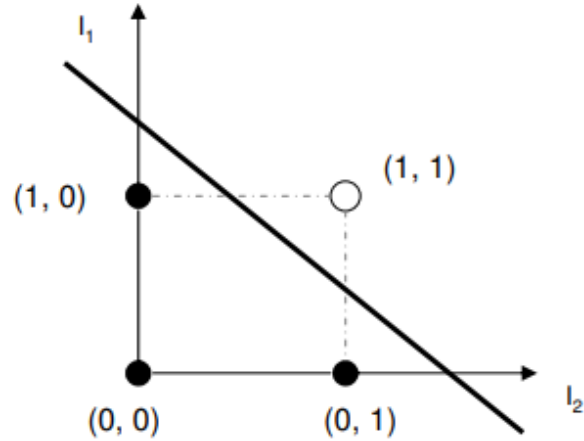


$$w_1x_1 + w_2x_2 + \dots + w_nx_n + b \geq k$$

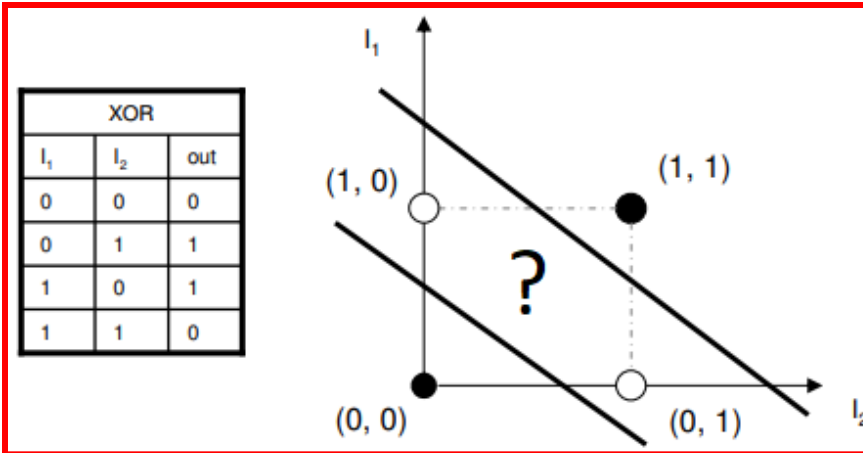
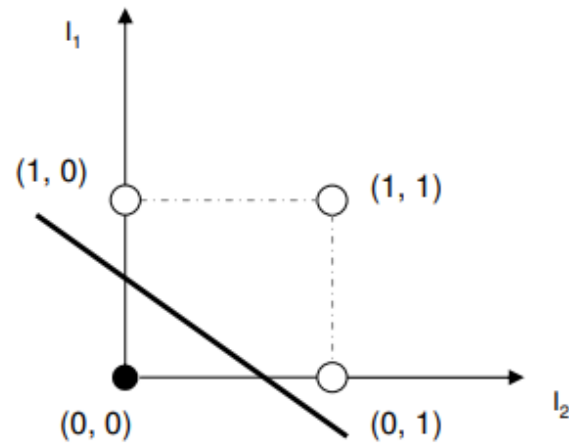
$$w_1x_1 + w_2x_2 + \dots + w_nx_n + b < k$$

Perceptron

AND		
I_1	I_2	out
0	0	0
0	1	0
1	0	0
1	1	1



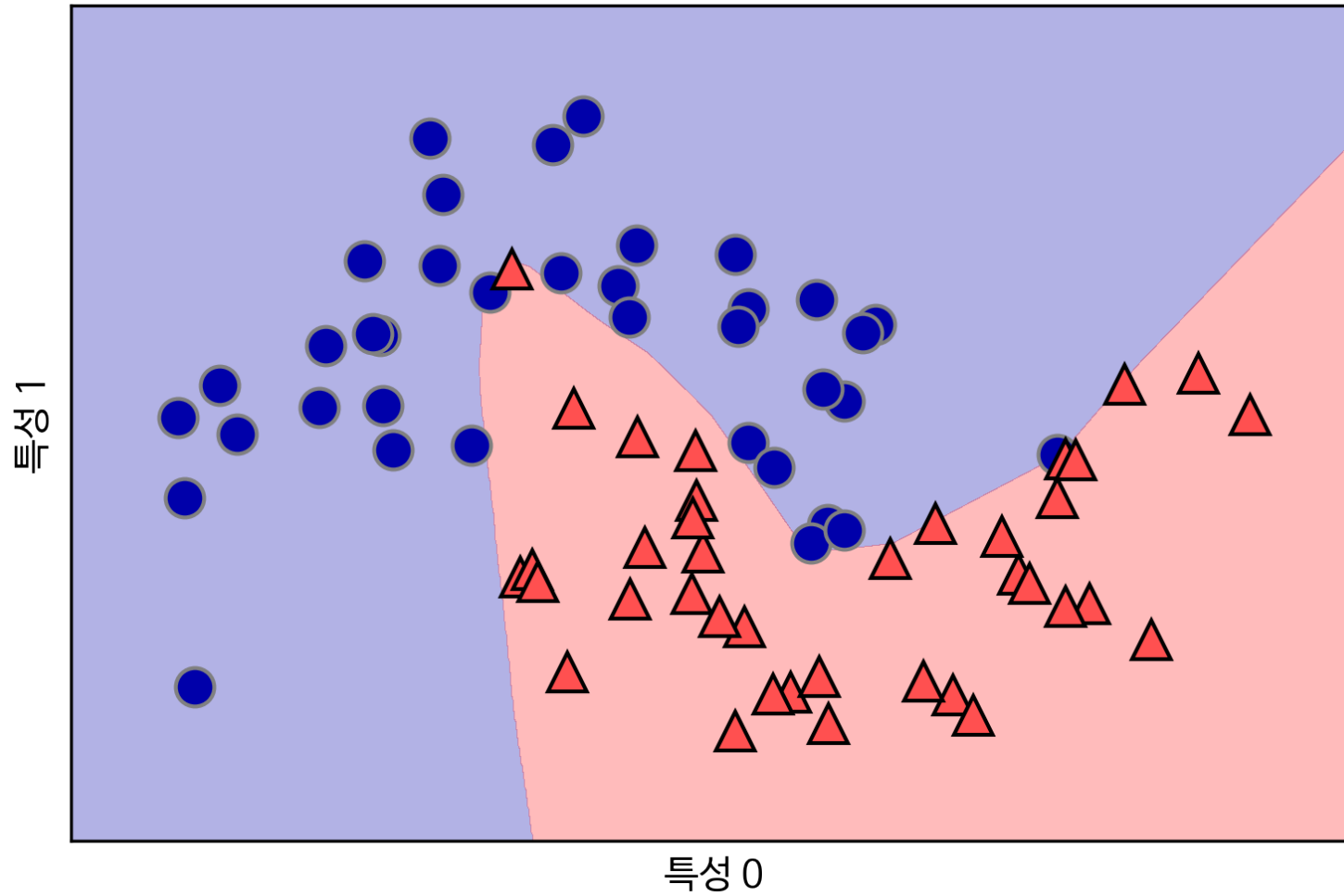
OR		
I_1	I_2	out
0	0	0
0	1	1
1	0	1
1	1	1



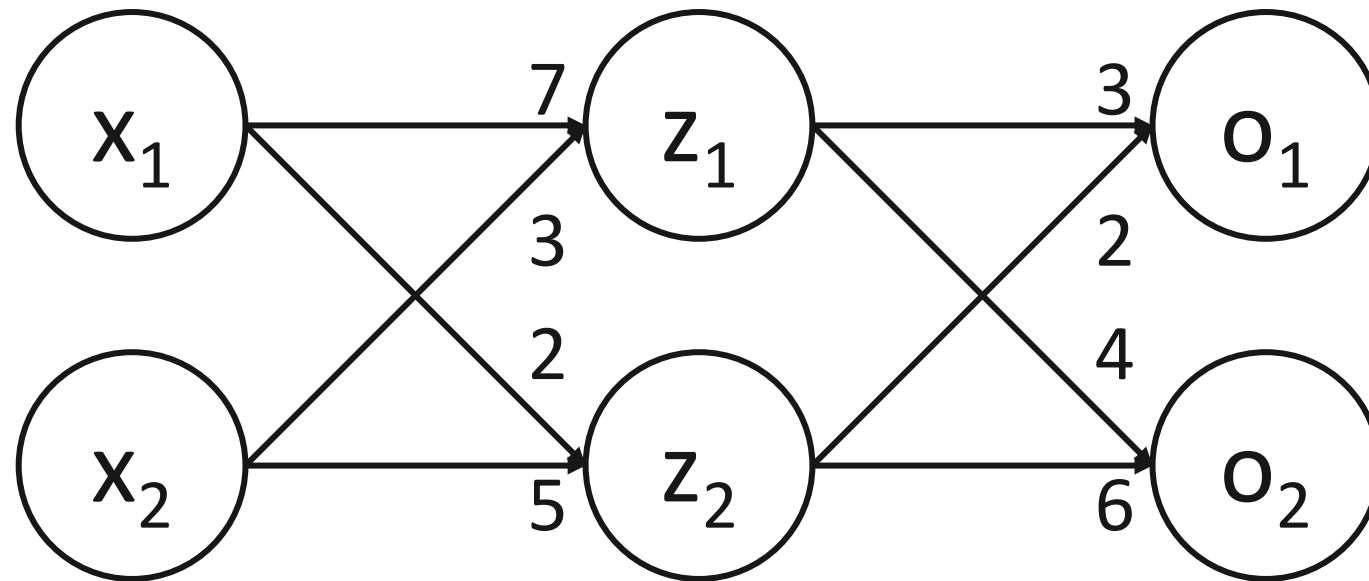
$$w_1 I_1 + w_2 I_2 + b \geq 0 \rightarrow 1$$

$$w_1 I_1 + w_2 I_2 + b < 0 \rightarrow 0$$

Multi-Layer Perceptron



Multi-Layer Perceptron



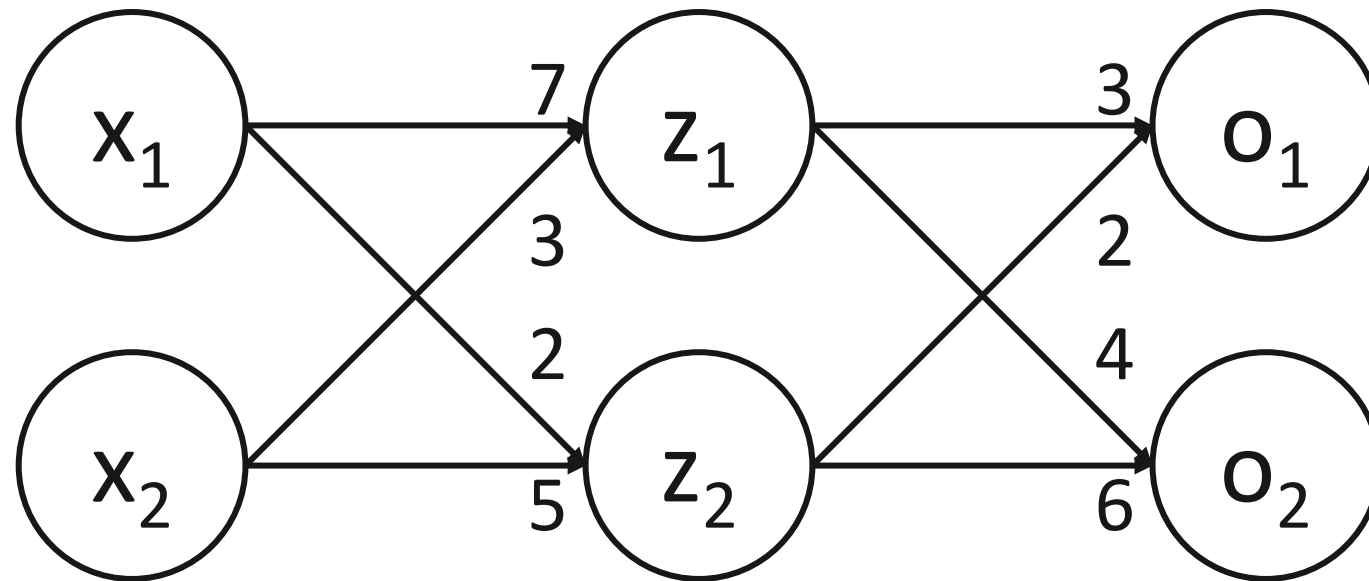
$$z_1 = 7x_1 + 3x_2$$

$$o_1 = 3z_1 + 2z_2$$

$$z_2 = 2x_1 + 5x_2$$

$$o_2 = 4z_1 + 6z_2$$

Multi-Layer Perceptron



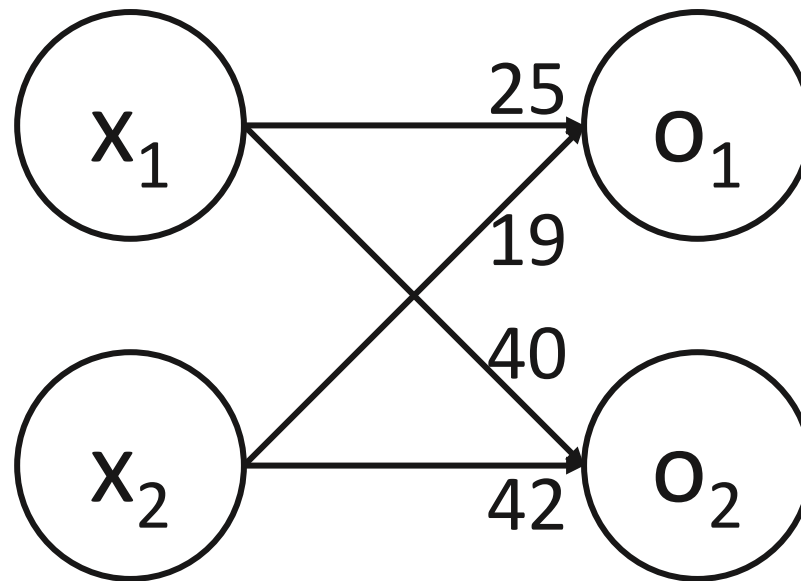
$$z_1 = 7x_1 + 3x_2$$

$$o_1 = 3z_1 + 2z_2 = 25x_1 + 19x_2$$

$$z_2 = 2x_1 + 5x_2$$

$$o_2 = 4z_1 + 6z_2 = 40x_1 + 42x_2$$

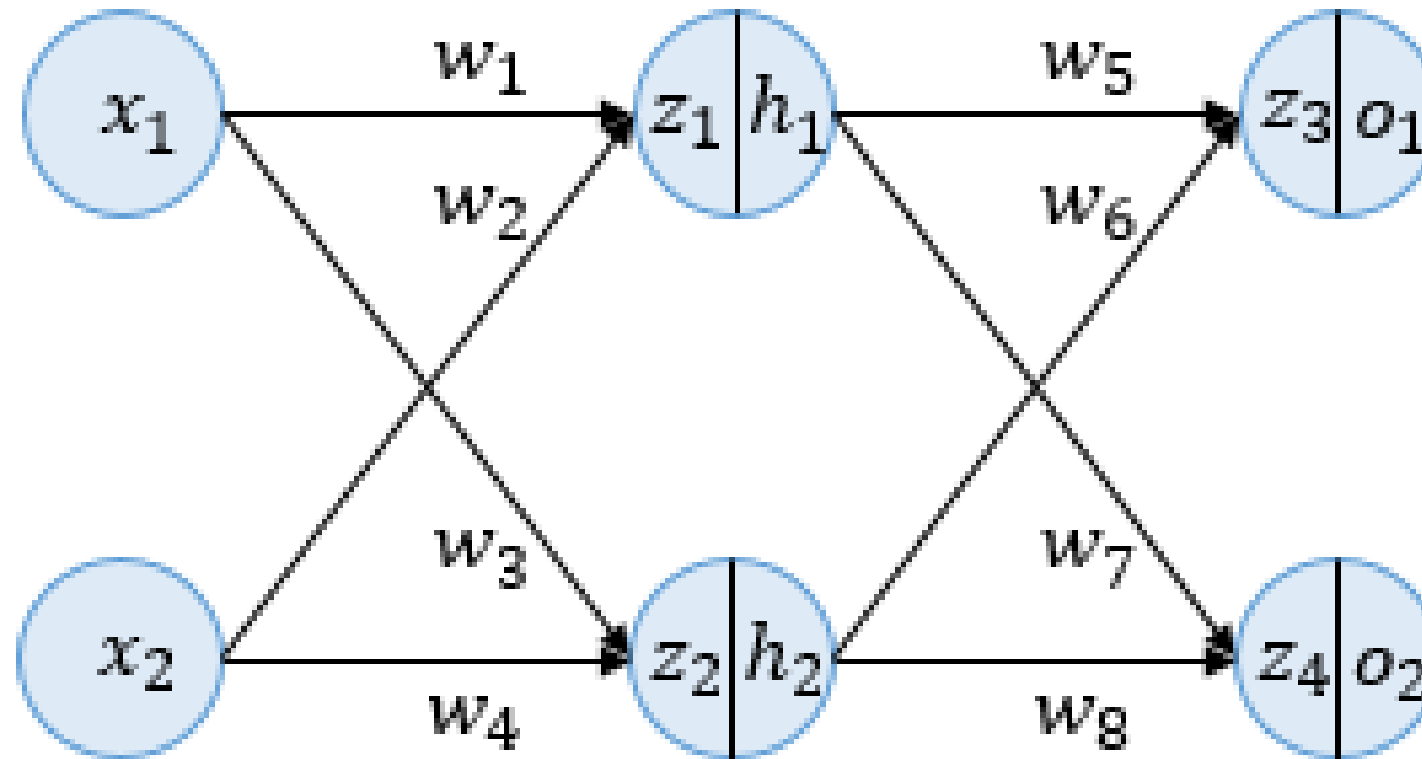
Multi-Layer Perceptron



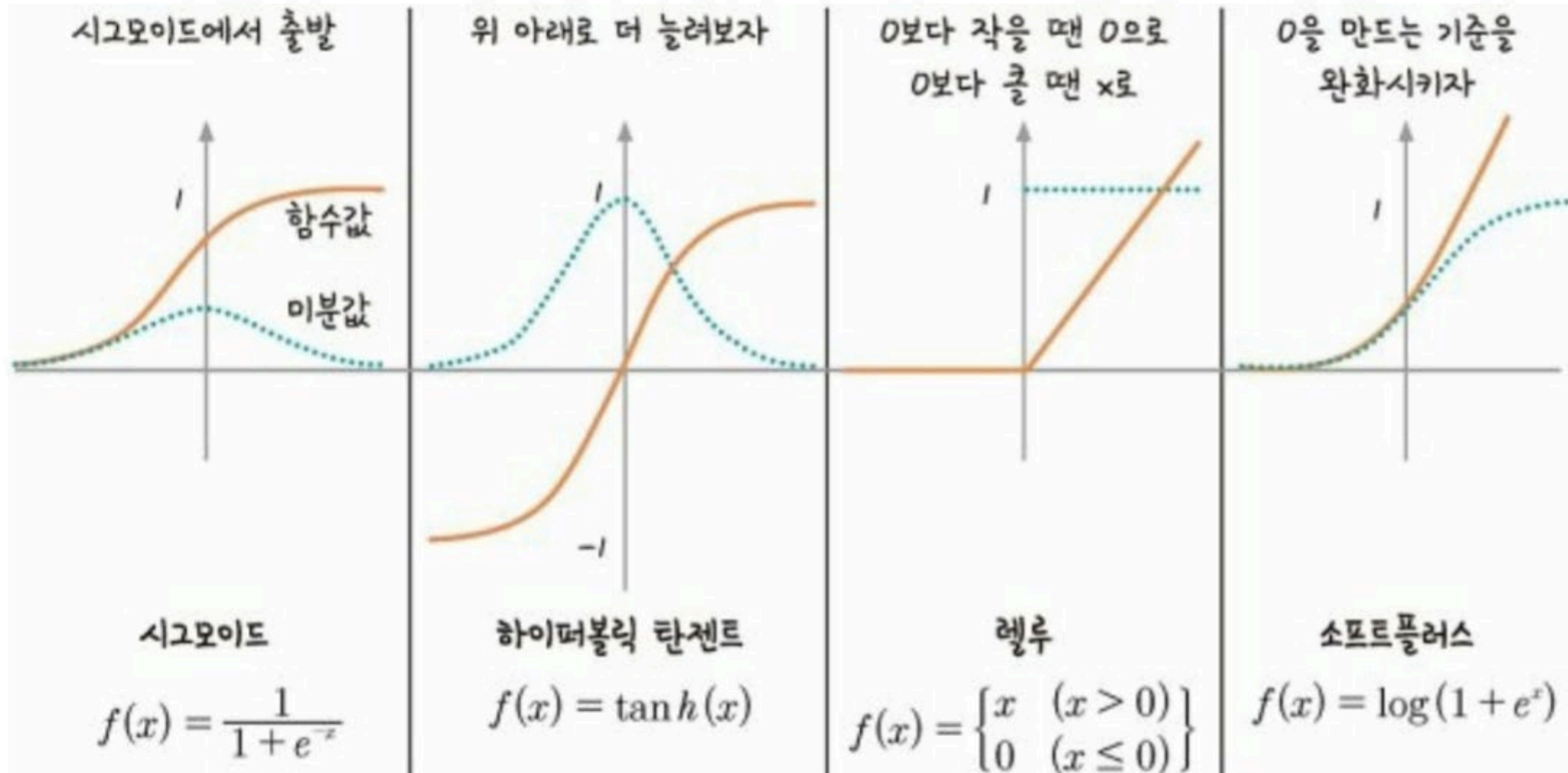
$$o_1 = 25x_1 + 19x_2$$

$$o_2 = 40x_1 + 42x_2$$

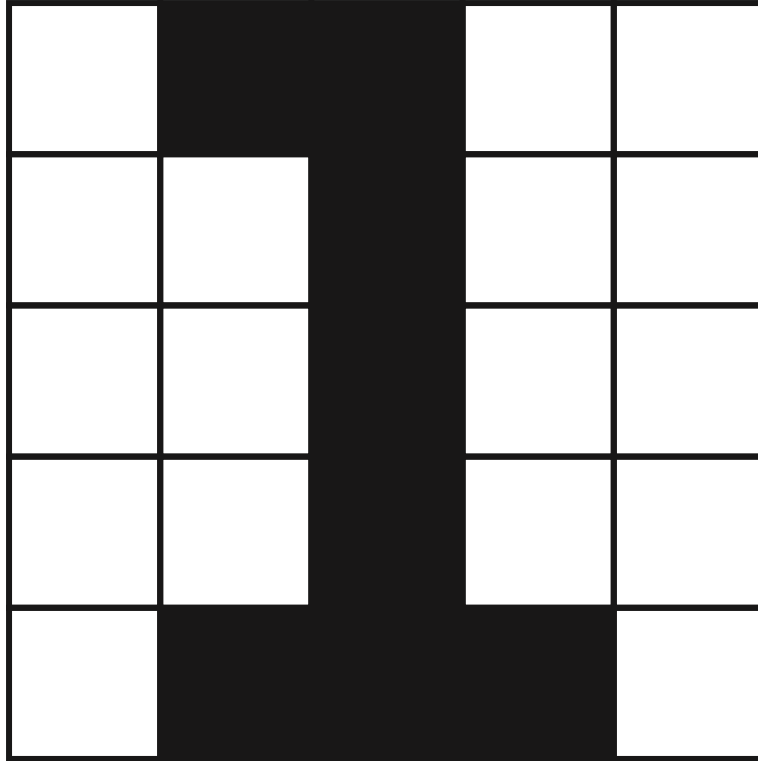
Multi-Layer Perceptron



Activation Function

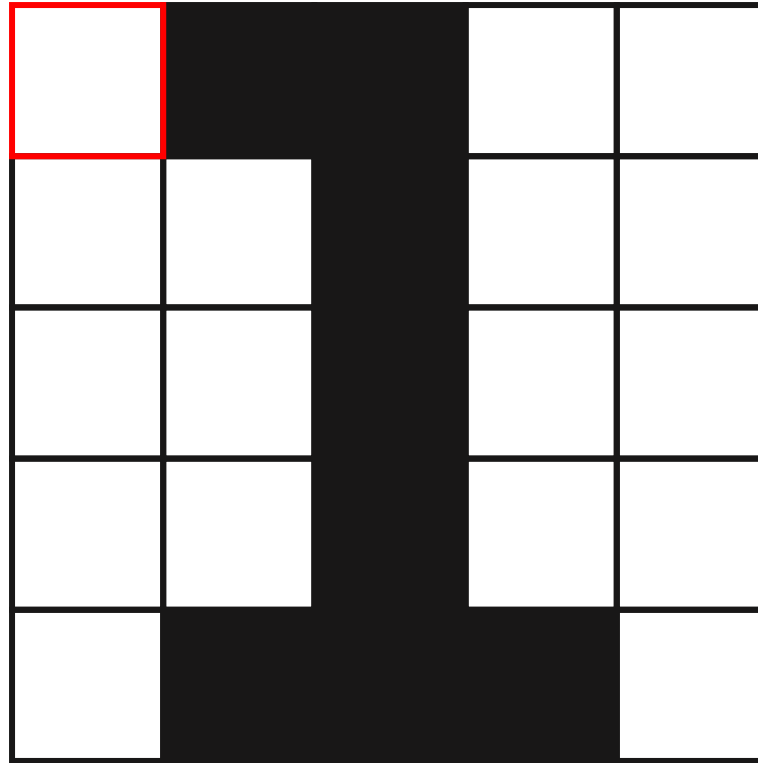


불필요한 연산



$$z_1 = w_1x_1 + w_2x_2 + \cdots + w_{25}x_{25} + b$$

불필요한 연산

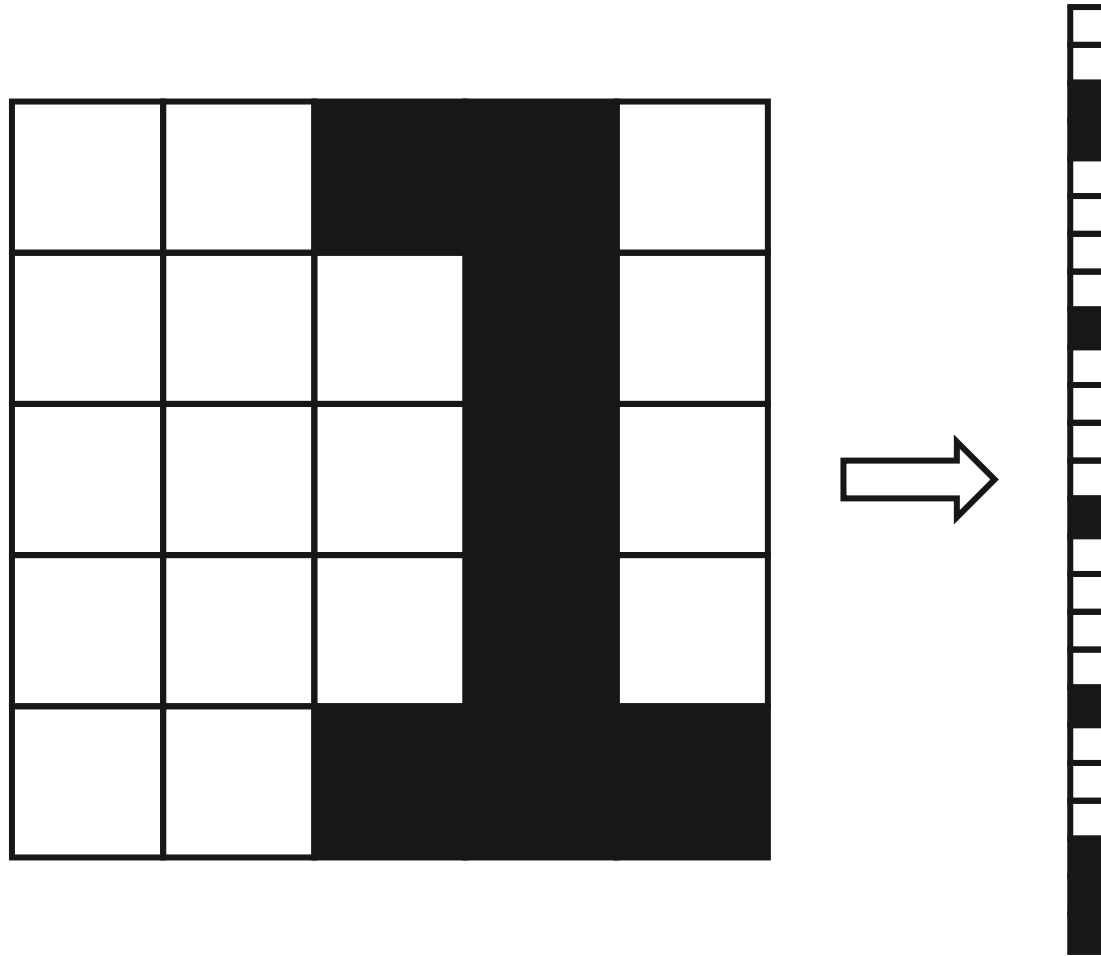


$$z_1 = w_1x_1 + w_2x_2 + \cdots + w_{25}x_{25} + b$$

$w_1?$



데이터 형상 무시

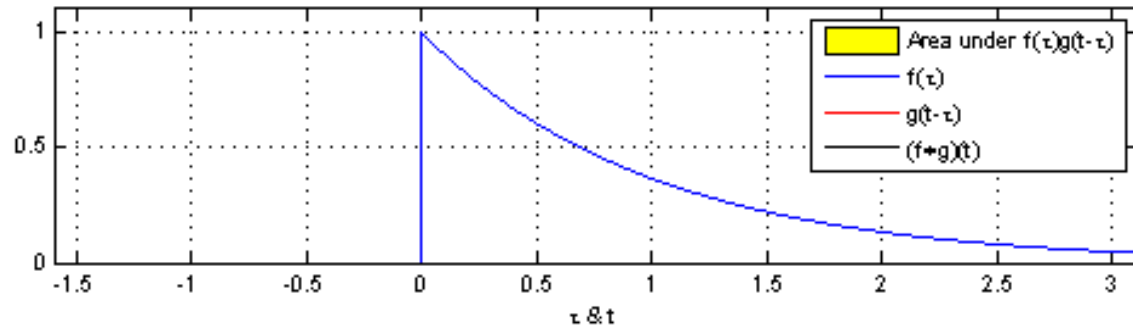


데이터 형상 무시

- 괜찮아요 한글 어순은 틀려도 다 알아들어요
- 한글 어순은 틀려도 괜찮아요 다 알아들어요
- 다 알아들어요 괜찮아요 한글 어순은 틀려도
- 전부 다 알아들어요 괜찮아요 한글 어순은 틀려도

Convolution

- Convolution (합성곱)
 - 두 함수를 곱해서 합한다
 - $(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)d\tau$



Convolution

1	2	3	4	5
2	1	0	1	2
3	0	1	1	0
1	4	1	1	2
2	1	1	0	0

입력

*

1	0	1
1	0	1
0	1	0

커널

=

출력

Convolution

1	2	3	4	5
2	1	0	1	2
3	0	1	1	0
1	4	1	1	2
2	1	1	0	0

입력

*

1	0	1
1	0	1
0	1	0

커널

=

6		

출력

Convolution

1	2	3	4	5
2	1	0	1	2
3	0	1	1	0
1	4	1	1	2
2	1	1	0	0

입력

*

1	0	1
1	0	1
0	1	0

커널

=

6	9	

출력

Convolution

1	2	3	4	5
2	1	0	1	2
3	0	1	1	0
1	4	1	1	2
2	1	1	0	0

입력

*

1	0	1
1	0	1
0	1	0

커널

=

6	9	11

출력

Convolution

1	2	3	4	5
2	1	0	1	2
3	0	1	1	0
1	4	1	1	2
2	1	1	0	0

입력

*

1	0	1
1	0	1
0	1	0

커널

=

6	9	11
10		

출력

Convolution

1	2	3	4	5
2	1	0	1	2
3	0	1	1	0
1	4	1	1	2
2	1	1	0	0

입력

*

1	0	1
1	0	1
0	1	0

커널

=

6	9	11
10	4	4
7	7	4

출력

커널

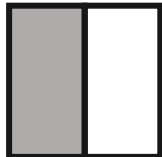
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0

*

1	0	-1
1	0	-1
1	0	-1

=

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0



커널

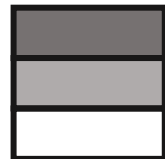
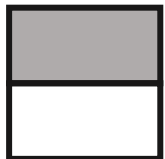
10	10	10	10	10	10
10	10	10	10	10	10
10	10	10	10	10	10
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

*

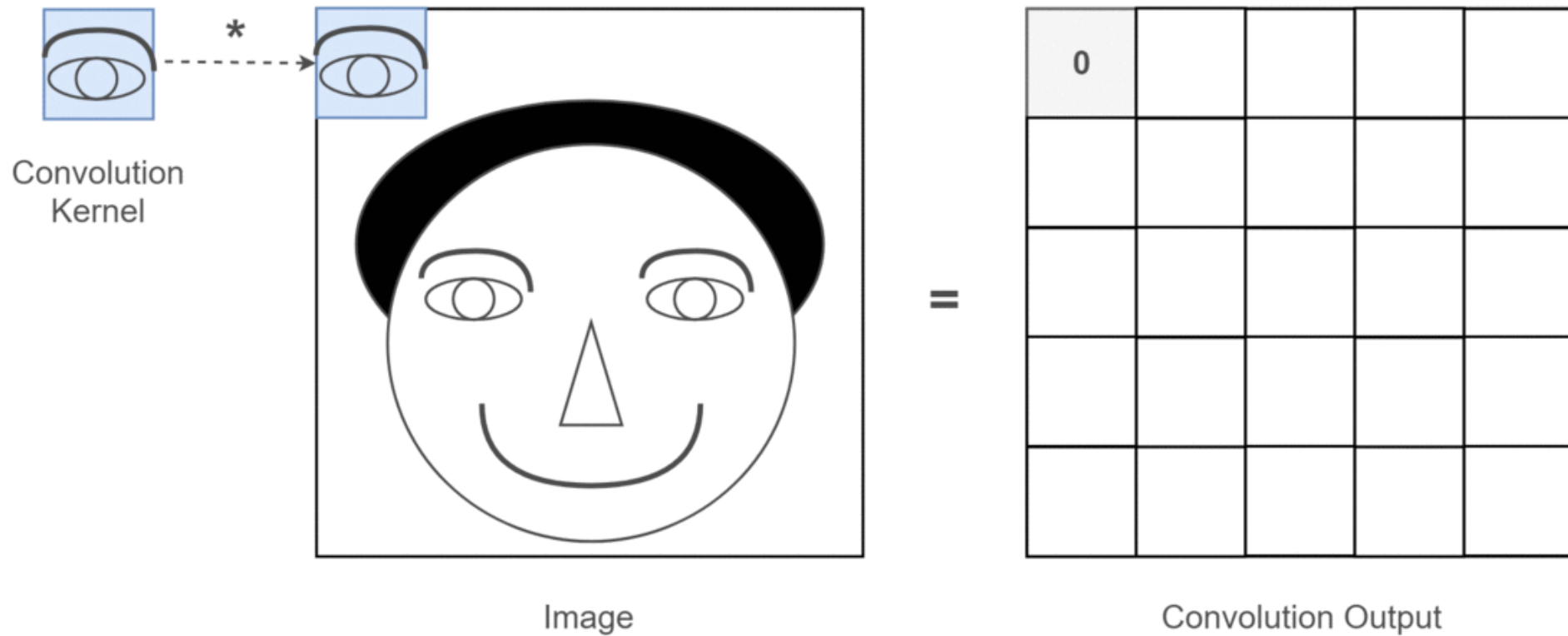
1	1	1
0	0	0
-1	-1	-1

=

0	0	0	0
30	30	30	30
30	30	30	30
0	0	0	0

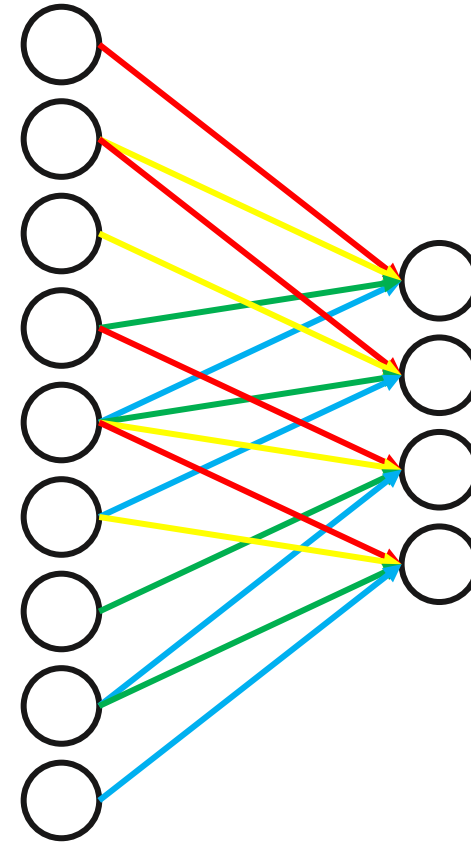


커널



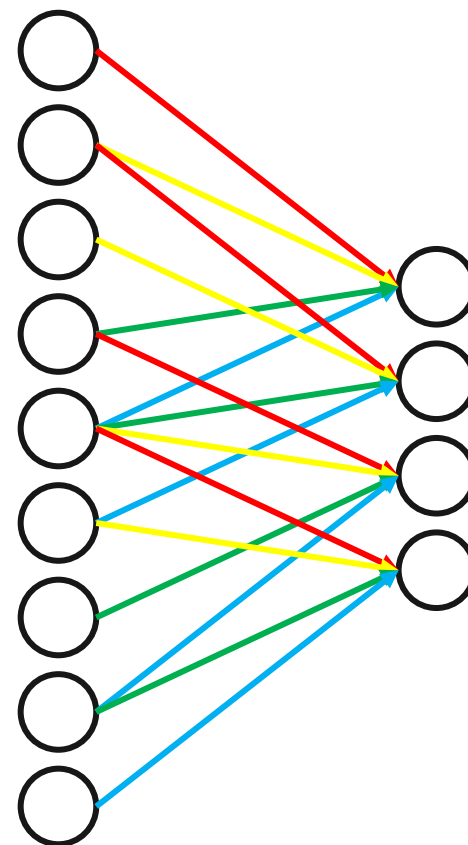
Convolution Layer

- Convolution 커널을 가중치로서 학습하는 Layer
 - AI 모델이 스스로 유의미한 이미지 패턴을 탐색
- $(3 \times 3 \text{ 이미지}) * (2 \times 2 \text{ 커널}) = (2 \times 2 \text{ 특성 맵})$



Convolution Layer

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 - AI 모델이 스스로 유의미한 이미지 패턴을 탐색
- $(3 \times 3 \text{ 이미지}) * (2 \times 2 \text{ 커널}) = (2 \times 2 \text{ 특성 맵})$
- 결국 MLP Layer의 일종
 - 모든 Node를 연결하지 않음
 - 가중치를 공유함

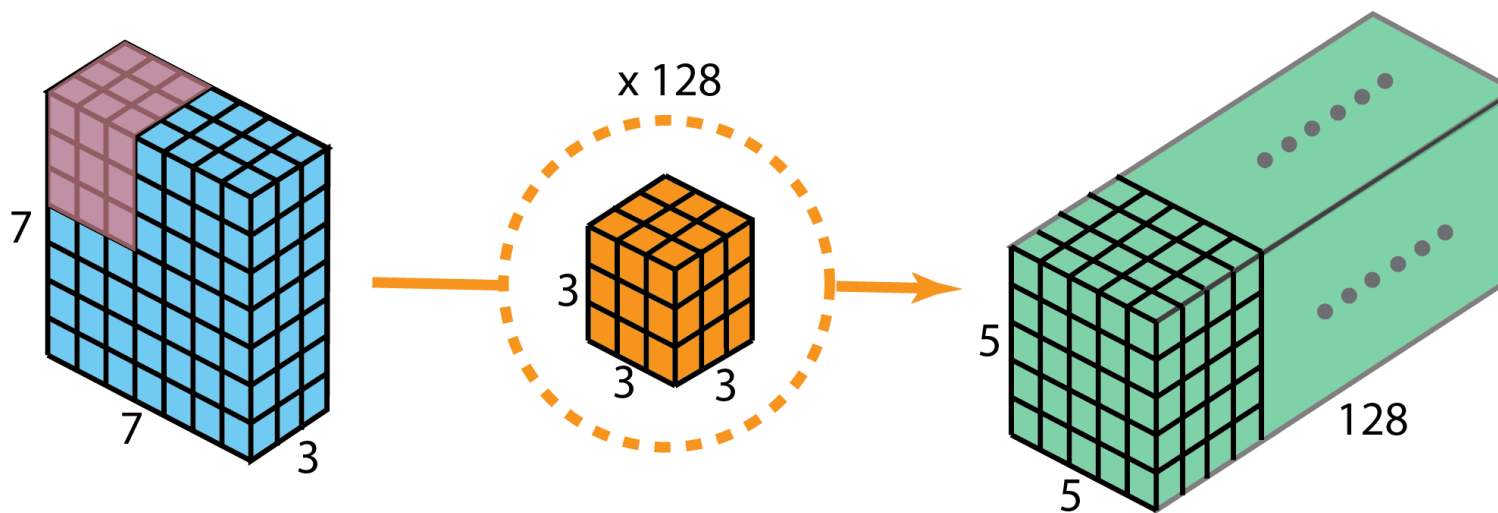


Convolution Layer

- 채널
 - 실제 이미지는 3차원 형태의 데이터 (채널 x 가로 x 세로)
 - MNIST 데이터셋 (흑백) – 1 x 28 x 28 이미지
 - CIFAR10 데이터셋 (컬러) – 3 x 32 x 32 이미지

Convolution Layer

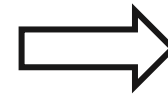
- 채널
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 - MNIST 데이터셋 (흑백) – 1 x 28 x 28 이미지
 - CIFAR10 데이터셋 (컬러) – 3 x 32 x 32 이미지
- $(C_I \times I_W \times I_H \text{ 이미지}) * (C_I \times K_W \times K_H \text{ 커널 } C_O \text{ 개}) = (C_O \times O_W \times O_H \text{ 특성 맵})$



Pooling Layer

- Pooling
 - 특정 영역 내에서 대표값을 선정하는 연산
- 2 x 2 Max Pooling Layer

5	2	3	4
2	1	0	1
3	7	9	5
1	4	1	8

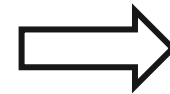


5	

Pooling Layer

- Pooling
 - 특정 영역 내에서 대표값을 선정하는 연산
- 2 x 2 Max Pooling Layer

5	2	3	4
2	1	0	1
3	7	9	5
1	4	1	8

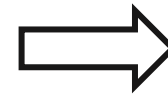


5	4

Pooling Layer

- Pooling
 - 특정 영역 내에서 대표값을 선정하는 연산
- 2 x 2 Max Pooling Layer

5	2	3	4
2	1	0	1
3	7	9	5
1	4	1	8

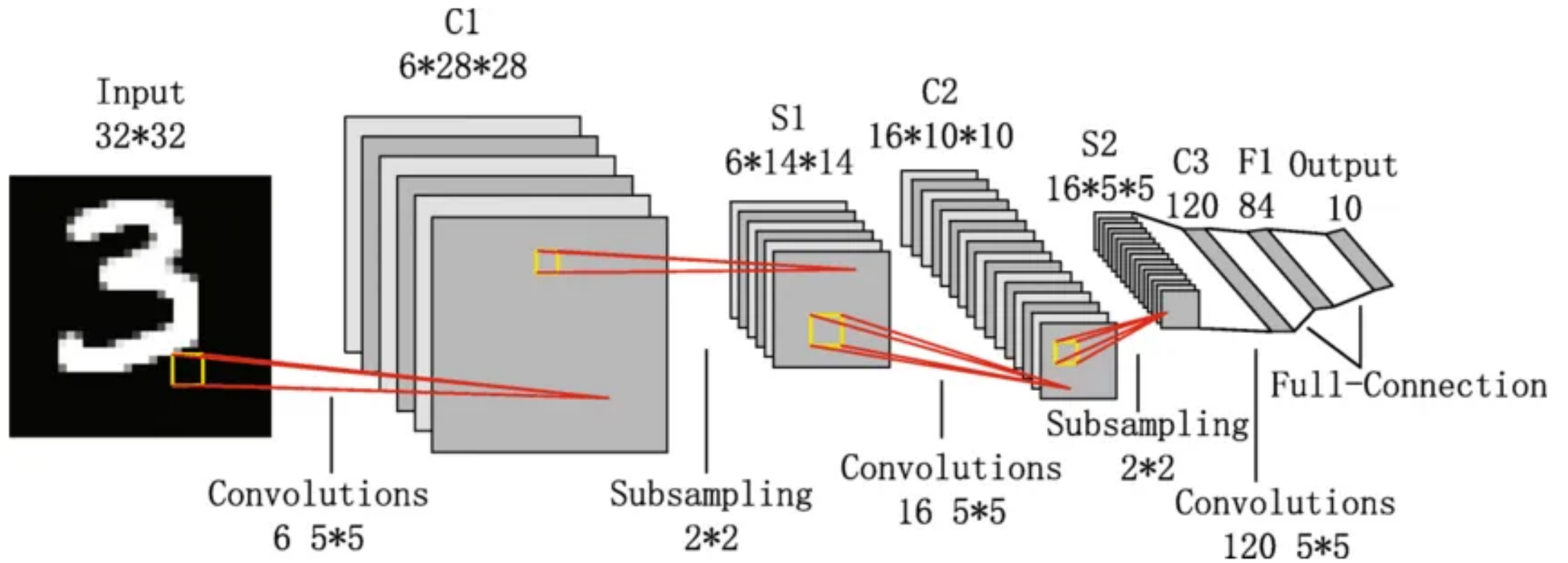


5	4
7	9

Pooling Layer

- 이미지는 많은 픽셀이 존재하기에 특성 맵의 크기를 축소하여 연산량을 감소
- Convolution Layer와 달리 학습할 가중치가 없음
- 이미지 속 객체의 위치나 형태가 약간 변형되어도 동일한 특성으로 인식 가능
- Pooling 방식은 다양
 - Max, Min, Average 등
 - 대부분의 경우 채널 별로 수행

Convolutional Neural Network



LeNet-5 at a glimpse

Convolution → Pooling → Convolution → Pooling → ... → MLP

심화

- Stride
- Padding
- Dilation
- Dropout
- Batch Normalization
- Residual Connection
- Bottleneck Layer